# Progressive Hypergraph Structure Learning for Fault Diagnosis of Industrial Robots

Tao Wang, Weize Xu, Chong Chen, Zhuowei Wang, and Zhuyun Chen

***Abstract*—The safe operation of industrial robots is crucial for maintaining productivity and ensuring product quality. However, the complexity of industrial robots has led to more diverse and unpredictable faulty modes, which poses significant challenges for accurate fault diagnosis. The existing deep learning algorithms often struggle with the capture of high-level features and the resistance to noise, resulting in inaccurate fault diagnosis. To address these limitations, this study proposes a new Hypergraph Neural Network (HGNN) for fault diagnosis. The proposed algorithm integrates three key modules: Feature Convolution, Progressive Hypergraph Structure Learning (PHSL), and HGNN classifier. The Feature Convolution module reduces noise and enhances data quality by aggregating both local and global information. The PHSL module refines the hypergraph structure through a progressive strategy, which can adaptively learn the complex and high-order correlations. Finally, the HGNN module performs robust fault diagnosis using the optimized hypergraph structure. Experimental results on a six-axis industrial robot fault diagnosis dataset demonstrate that the proposed algorithm achieves higher accuracy and robustness compared to state-of-the-art benchmark algorithms. The outcome of this study can bring tangible benefits to the safe operation of industrial robots. The source code of this study is available at: https://github.com/Wade914/PHSL**

***Index Terms*—Hypergraph Neural Network, Fault Diagnosis, Hypergraph Structure Learning, Industrial Robots.**

## I. Introduction

THE rapid advancement of industrial automation and intelligent systems has significantly increased the complexity of industrial equipment, making fault diagnosis a critical task for ensuring operational efficiency and safety [**10100629**]. Industrial robots have gained increasing attention in recent years. In particular, industrial robots, which are widely used in manufacturing processes, require precise and timely fault diagnosis to prevent costly downtime and maintenance issues [**chen2023deep**]. As a type of mechatronic device, it is challenging to identify the faulty parts in the early faulty stage since the incipient faulty patterns are hard to extract under noisy operation conditions [**chen2025multi**]. Hence, it is essential to investigate advanced algorithms that can effectively capture faulty patterns and be resistant to noise.

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Recent developments in deep learning have shown great promise in addressing the challenges of high-level feature extraction and noise resistance. Among these, Graph Neural Networks (GNNs) have gained attention due to their ability to model relationships between entities represented as nodes in a graph. Despite their success, GNNs face limitations when dealing with higher-order interactions and large-scale datasets [**pistilli2023graph**]. Advanced algorithms such as Graph Convolutional Network (GCN) [**kipf2016semi**] Graph Attention Network (GAT) [**velivckovic2017graph**], and Interaction-Aware GNN (IAGNN) [**chen2021interaction**] have shown merits in fault diagnosis. However, these GNN algorithms are struggling to mine the hidden patterns in time series. To overcome these shortcomings, HGNNs have emerged as a powerful tool, capable of capturing multi-way relationships among multiple nodes simultaneously [**ke2023time**]. However, the obtained multi-way relationships may contain a certain level of noise, which impedes the accurate diagnosis of HGNNs.

This paper proposes a fault diagnosis algorithm specifically designed for industrial robots, which can leverage the strengths of HGNN and lower the impact of noise from hypergraphs. The proposed algorithm consists of a feature convolution module, a PHSL module, and an HGNN module. The main contributions of this work are threefold: (1) a feature convolution module is designed that combines depthwise and pointwise convolutions to efficiently aggregate local and global information, which can reduce redundancy while preserving essential details. (2) A PHSL module is proposed to dynamically adjust the hypergraph structure through a progressive node splitting and refinement strategy, which helps the model to capture high-order relationships and be resistant to noise. (3) Extensive experiments on a real-world industrial robot dataset validate the effectiveness of our method, which shows merits in the diagnosis accuracy compared to existing benchmarking algorithms.

## II. Literature Review

### A. Time Series Feature Extraction for Fault Diagnosis

The accurate extraction from industrial robot time-series data remains challenging due to complex variable interactions and high-order correlations. Traditional frequency-domain methods struggle with non-smooth signals. Yemets et al. [**yemets2025time**] introduced a FFT-based transformer model for time series forecasting, which transformes data into the frequency domain and incorporating amplitude and phase information. However, FFT's inherent limitations in time-frequency localization persist [**gupta2019feature**]. Early improvements like the Short-Time Fourier Transform (STFT) attempted time-frequency localization through sliding windows. For instance, Tu et al. [**tu2024stft**] developed the STFT-TCAN model, combining STFT with Temporal Convolutional Networks and attention mechanisms to extract time series features. More recently, Lei and Wang [8] developed DWT-TimesNet, leveraging wavelet transforms for multi-resolution analysis and dynamic thresholding to isolate valid cycles.

While traditional methods like FFT [**yemets2025time**] and STFT [**tu2024stft**] struggle with nonlinear non-stationary signals, deep learning architectures enable adaptive feature learning through temporal convolution and self-attention. Convolutional neural networks (CNN) have been widely adopted to address the limitations of rigid spectral transforms. For instance, Cao et al. [**cao2024remaining**] demonstrated that TCN-Transformer hybrid models suppress noise in rolling bearing degradation indices through dilated convolutions. However, CNN-based approaches often fail to capture long-range dependencies critical for industrial data. To bridge this gap, Fang et al. [**fang2024convolution**] proposed a convolution-augmented Transformer combining local convolutional features with global attention, showing efficacy in aircraft bearing diagnosis. Khaniki et al. [**khaniki2023enhancing**] validating the efficacy of LSTM-attention mechanisms in enhancing discriminative fault signature identification for induction motors.

Despite these advancements, deep learning models still face challenges in feature extraction of noisy data. Recent studies have introduced patch-based representations to enhance feature extraction. For example, Gong et al.[**gong2023patchmixer**] proposed PatchMixer using depthwise separable convolutions for local patch processing. This approach inspired Wang et al. [**wang2025multi**] to develop MP-MixerNet, incorporating dual-scale modeling and dynamic instance normalization to handle distribution shifts. Building on these foundations, Cao et al. [**cao2025mspatch**] advanced multi-scale adaptability through MSPatch, which decomposes time series into hierarchical patches with dynamic weight. While these methods improve feature extraction for fault diagnosis efficiency, they remain constrained by interrelated limitations: neglect of cross-patch heterogeneous interactions, rigidity in adapting to dynamic fault evolution, and an inability to explicitly model high-order variable correlations.

### *B. Recent Advance of Graph Neural Networks*

GNNs have become a cornerstone in modeling relational data, which offers a natural way to represent interactions between entities as nodes and edges in a graph. Traditional GNNs, such as GCN [**kipf2016semi**] and GAT [**velivckovic2017graph**], have been adopted for fault diagnosis tasks due to their ability to capture pairwise relationships. Chen et al. [**chen2021interaction**] introduced the Interaction-Aware GNN, which represents sensor signals as a heterogeneous graph with adaptive edge weights via attention mechanisms, enabling dynamic fault propagation modeling. While IAGNN processes subgraphs for distinct interactions, it remains confined to pairwise relationships. Babu et al. [**babu2024dynamic**] extended this by integrating GNNs with LSTMs to model spatial-temporal dependencies in wind turbine data. Zhu et al. [**zhu2025dual**] proposed a Dual-Contrastive Multiview GAT that combines multi-view contrastive learning with dynamic adjacency matrix optimization, which improves robustness to distribution shifts.

Despite these improvements, existing GNN variants fail to explicitly represent multi-component interactions due to their pairwise-centric design. Such limitations highlight the need for HGNNs that generalize graph structures to model high-order interactions through multi-node hyperedges. Yan et al. [**yan2024multisensor**] pioneered multisensor hypergraph fusion with adaptive hyperedge weighting to jointly model local component dynamics and global system dependencies. Building on this, Xia et al. [**xia2022residual**] introduced Res-HGCN, which integrates residual connections with physics-based hypergraph initialization to enhance interpretability. To address noise interference and small-sample challenges, Lei et al. [**lei2023signal**] proposed WC-HGCN, a dual-channel framework combining wavelet convolution for noise-robust time-frequency analysis and hypergraph convolution for dynamic sensor correlation mining. Chen et al. proposed a multi-scale graph pyramid attention network for robotic fault diagnosis. Multi-scale pyramid attention mechanism was adopted to extract the faulty patterns in different scales so to achieve better fault diagnosis accuracy.

While HGNNs expertise on modeling high-order relationships, their performance critically depends on initial hypergraph quality. In real-world applications, the initial hypergraph structure is often noisy or incomplete, containing erroneous connections or missing critical relationships. To address this issue, Zhao et al. [**zhao2023dynamic**] proposed Dynamic Hypergraph Structure Learning (DyHSL), which dynamically constructs hyperedges from node representations to capture non-pairwise spatiotemporal interactions. Bollengier et al. [**bollengier2024dynamic**] further adopted this paradigm in multimodal disease diagnosis, introducing a bi-clustering mechanism to generate adaptive hypergraphs that model variable-sized node group correlations. Extending these ideas, Ju et al. [**ju2024hypergraph**] developed HyperMatch with vertex-hyperedge alternating attention, iteratively refining hypergraph structures through cosine similarity-based feature grouping and dual-view optimization.

In summary, accurate extraction of key features from time series data is fundamental for effective fault diagnosis. Traditional methods, such as FFT and STFT, struggling with non-linear nonstationary signals and lack adaptability in dynamic environments. While deep learning algorithms have advanced feature extraction through adaptive temporal convolution and self-attention mechanisms, they often neglect critical long-range temporal dependencies. Existing GNNs, despite their capability to model pairwise relationships between nodes, fall short in representing multi-component interactions due to their binary-centric design. HGNNs have emerged as a promising tool capable of simultaneously capturing multi-way relationships among multiple nodes, addressing some of the limitations of traditional GNNs. However, the performance of HGNNs heavily relies on the quality of the underlying hypergraph structure, which can be noisy or incomplete in real-world applications. Hence, it is essential to investigate a new HGNN that can automatically adjust its structure and be resistant to noise during fault diagnosis modeling.

## III. Methodology

In industrial robotic operations, the need to capture high-order features and the presence of noise pose great challenges for accurate fault diagnosis. Therefore, the proposed method aims to these limitations by integrating three key modules: the features convolution module, the progressive hypergraph structure learning module, and the hypergraph neural network module. The overall architecture of the proposed model is illustrated in Fig. 1. The features convolution module denoises raw equipment signals via patch-based local-global information aggregation, achieving dimensionality reduction while preserving discriminative features. The progressive hypergraph structure learning module refines the hypergraph structure via a progressive strategy, ensuring that the model can adaptively learn the complex and high-order correlations among the data. Finally, the hypergraph neural network module performs the fault diagnosis task.

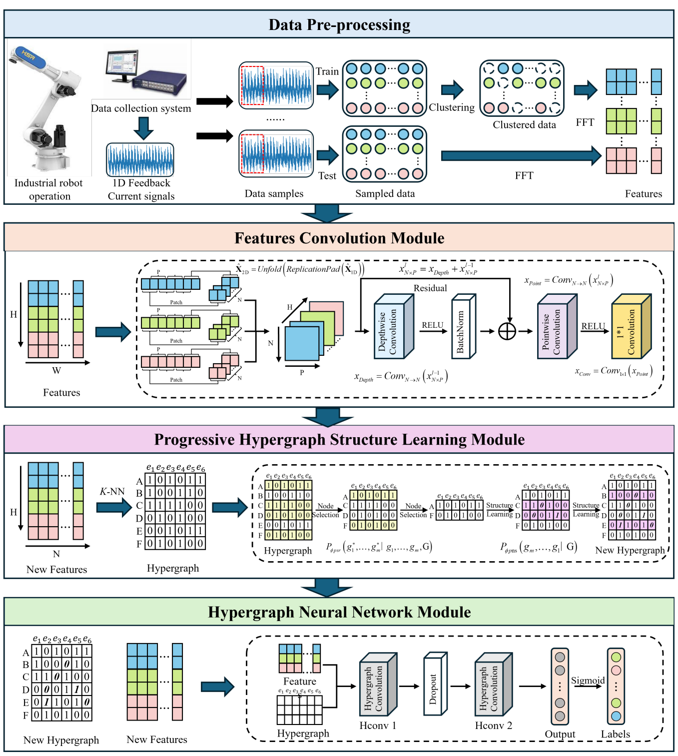



Fig. 1: Overall flowchart of the proposed algorithm.

### A. Feature Convolution

The input time series data are segmented using a sliding window with a length of P and a step of S. This process transforms the original univariate time series data into two-dimensional patches while preserving the original temporal relationships between the data points.

$$\hat{\mathbf{X}}_{2\mathrm{D}} = Unfold\left(ReplicationPad\left(\hat{\mathbf{X}}_{1\mathrm{D}}\right)\right) \\ \text{size} = P, \text{step} = S \tag{1}$$

#### 1) Depthwise Convolution:

The Depthwise Convolution is applied to capture potential periodic patterns within the temporal patches via convolving patches that share the same spatial positions. Specifically, the Depthwise Convolution employs a specialized form of grouped convolution where the number of groups is equal to the number of patches, denoted as N. To achieve a broader receptive field, the kernel size is set to a relatively large value. Each of the N patches in the input feature map undergoes a separate convolution with a distinct kernel, resulting in N feature maps. The equation below illustrates the process of the depthwise convolution where the input data $x_{N\times D}$ in layer $l-1$ is processed through the depthwise convolution in layer $l$ , $\sigma$ denotes an activation function. In this case, we employ the RELU activation function.

$$x^{l}_{N\times P} = BatchNorm\Big(\sigma\Big\{Conv_{N\to N}(x^{l-1}_{N\times P}, \\ \text{stride} = 7, \text{kernel\_size} = 7)\Big\}\Big) + x^{l-1}_{N\times P} \tag{2}$$

#### 2) Pointwise Convolution:

To effectively capture the correlations between patches, a pointwise convolution is applied following the depthwise convolution, which facilitates temporal interactions between the patches. The equation below illustrates the process of the pointwise convolution, where the input data $x_{N\times D}$ in layer $l$ is processed through the pointwise convolution in layer $l+1$, resulting in an output feature map with N channels.

$$x^{l+1}_{N\times P} = \sigma\{Conv_{N\to N}(x^{l}_{N\times P}, \\ \text{stride} = 1, \text{kernel\_size} = 1)\} \tag{3}$$

#### 3) Features Convolution:

Following the depthwise and pointwise convolutions, we apply a 1D convolutional layer to reduce the feature dimensions. This operation compresses the length of each patch, reducing the number of features while retaining essential information. The output feature maps are then passed through a max pooling layer, reducing the patch length from 2 to 1. Finally, the feature maps are flattened and serve as the input for the subsequent modules.

$$x^{output}_{N\times 1\times H} = Maxpool\Big(Conv_{1\times 1}(x^{input}_{N\times P\times H}, \\ \text{patch\_length}, 2, \text{kernel\_size} = 1)\Big) \tag{4}$$

$$x^{flat}_{H\times N} = Flatten\left(x^{output}_{N\times 1\times H}\right) \tag{5}$$

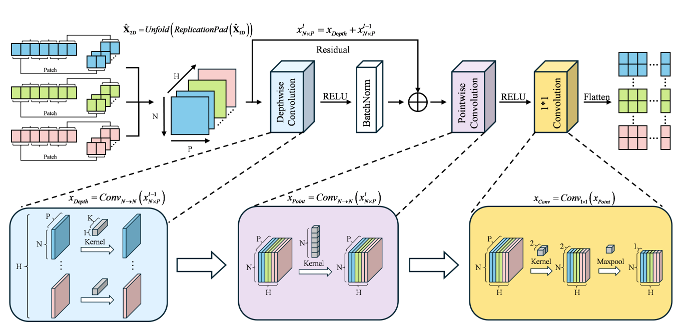



Fig. 2: Architecture of Feature Convolution.

### B. Progressive Hypergraph Structure Learning

Graph Structure Learning (GSL) is a crucial component in graph-based models, aiming to infer the underlying graph structure from the input data. Traditional GSL methods often rely on predefined or manually crafted graphs, which may not accurately reflect the true relationships among data points. To address this limitation, Hypergraph Structure Learning (HSL) is introduced, which extends the concept of GSL to hypergraph, which allows for the modeling of higher-order relationships among nodes. HSL enhances the capability of GSL by considering higher-order interactions.

To further enhance the learning of the hypergraph structure, we propose PHSL, which iteratively refines the hypergraph structure by incorporating feedback from the model's performance, ensuring that the learned structure is both accurate and robust. The progressive refinement process can be described by the following 2 major parts, progressive node splitting module and progressive refine module Traditional HSL methods do not consider the status of nodes and hyperedges, leading to a monotonous standard for evaluating all connections. This can result in indeterminacy and instability during the optimization process. PHSL, on the other hand, is status-aware, meaning it takes the importance and influence of different nodes and hyperedges into account. This is achieved through a progressive node splitting module (PNS) and progressive structure refining module (PSR).

$$P_{\phi}\left(\mathcal{G}^{*} \mid \mathcal{G}\right)=\prod_{i=1}^{m} P_{\phi_{i}}\left(g_{i}^{*} \mid g_{<i}^{*}, \mathcal{G}\right) \tag{6}$$

The optimized graph, denoted as $\mathcal{G}^*$ , can be viewed as a composite of m subgraphs. Each optimized subgraph $g_i^*$ is derived from the preceding subgraphs $g_1^*, g_2^*, \ldots, g_{i-1}^*$ to form a larger subgraph, in addition to the original graph $\mathcal{G}$ . Specifically, the input graph is progressively divided into m subgraphs $g_1, g_2, \ldots, g_m$ . The nodes in each subgraph are utilized to construct the corresponding optimized subgraph $g_i^*$ . Consequently, the graph structure learner can be decomposed into two primary modules: the Progressive Node Splitting Module and the Progressive Structure Refining Module.

$$\begin{aligned} P_{\phi}\left(\mathcal{G}^{*} \mid \mathcal{G}\right) &= P_{\phi}\left(g_{1}^{*}, \ldots, g_{m}^{*} \mid \mathcal{G}\right) \\ &= \sum_{(g_m,\ldots,g_1)\in\mathcal{G}} \Big[P_{\phi psr}\big(g_1^*, \ldots, g_m^* \mid g_1, \ldots, g_m, \mathcal{G}\big) \\ &\qquad \cdot P_{\phi \mathrm{pns}}\big(g_m, \ldots, g_1 \mid \mathcal{G}\big)\Big] \end{aligned} \tag{7}$$

*1) Progressive node splitting module:* The PNS module determines which nodes are included in each subgraph and the optimal order of subgraphs through a learnable progressive strategy. These selected nodes are referred to as "celebrity nodes." This process is repeated in subsequent layers, with increasing competition to narrow down the range of candidate nodes.

Specifically, the node selection is performed using a scoring function $HGNN_{rk}^{(l)}(\cdot)$ to evaluate the nodes in the previous subgraph $g_{i-1}$. Assuming that the number of nodes in the previous subgraph is x, and given a splitting ratio $r \in (0,1)$ , the top $r \times x$ nodes are selected for the next subgraph $g_i$ .

$$E^{(i)} = sigmoid\left(HGNN_{rk}^{(i)}\left(H^{(i)}, \mathcal{A}^{(i)}\right)\right), \tag{8}$$

$$idx^{(i)} = rank\left(E^{(i)}, r\right), \tag{9}$$

$$H^{(i+1)} = H^{(i)}\left[idx^{(i)}, :\right], \tag{10}$$

$$\mathcal{A}^{(i+1)} = \mathcal{A}^{(i)}\left[idx^{(i)}, :\right] \tag{11}$$

Where $E^{(i)}$ denotes the score vector for nodes in subgraph $g_i$, the operation $rank(\cdot)$ returns the indices of the top r proportion of the highest values in $E^{(i)}$.

*2) Progressive refine module:* Following the PSS module, the PSR module further optimizes the hypergraph structure. The PSR module inverts the pyramid-like architecture generated by the PSS module to create an inverted pyramid-like structure. Node position encoding is introduced to ensure that the recovered nodes are aware of their position within this architecture. The hypergraph structure is then refined progressively, which ensures that node connections are adjusted based on their global significance. This process enhances the entire structure refinement, making it more accurate and robust.

$$Z'^{(l)} = HGNN_{rf}\left(H^{(l)}, \mathcal{A}^{*(l)}\right), \tag{12}$$

$$Z^{(l)} = MLP_{\text{feat}}\left(concat\left(Z'^{(l)}, \mathcal{V}^{(l)}\right)\right), \tag{13}$$

$$\mathcal{M}^{(l)} = sim\left(Z_i^{(l)}, Z_j^{(l)}\right), \tag{14}$$

$$\mathcal{A}^{*(l-1)} = Fuse\left(\mathcal{A}^{(l-1)}, \mathcal{M}^{(l)}, idx^{(l-1)}\right) \tag{15}$$

Where $\mathcal{V}^{(l)}$ denotes the position vector of nodes subgraph $g_i$ , the operation $sim(\cdot)$ is an embedding distance metric function. $\mathcal{M}^{(l)}$ represents the subgraph structure, and the operation $Fuse(\cdot)$ function fuses the corresponding structure in $\mathcal{M}^{(l)}$ and $\mathcal{A}^{(l-1)}$ . It is worth noting that the subgraph order here is in descending order, whereas it was in ascending order previously.

## IV. Experimental Setup

### A. Data Collection

In this study, a composite fault injection experiment was conducted to gather fault data from a six-axis industrial robot, wherein faulty reducers and motors were utilized to replace the original functioning components. Early-stage defects in the faulty components manifested as minimal oil leakage of reducer and slight abnormal noises from the motor that develop into serious mechanical breakdowns while remaining difficult to detect because their fault patterns are unclear.In this experiment, three-phase current signals were collected through a Hall element built into the motor driver and The third axis feedback current was used as input in this study. The modeling process required q-axis current measurements taken at 1Hz sampling frequency. The methodology enables researchers to collect data that precisely shows the initial stages of potential

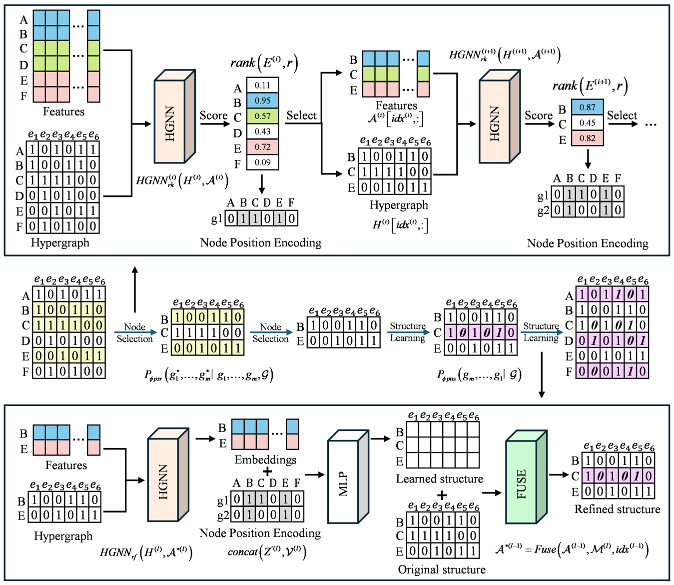



Fig. 3: Architecture of the proposed PHSL.

faults to help identify faults at an early stage. The fault data specifically encompassed the feedback current of the third axis. These current signals are depicted in Fig 4. The dataset

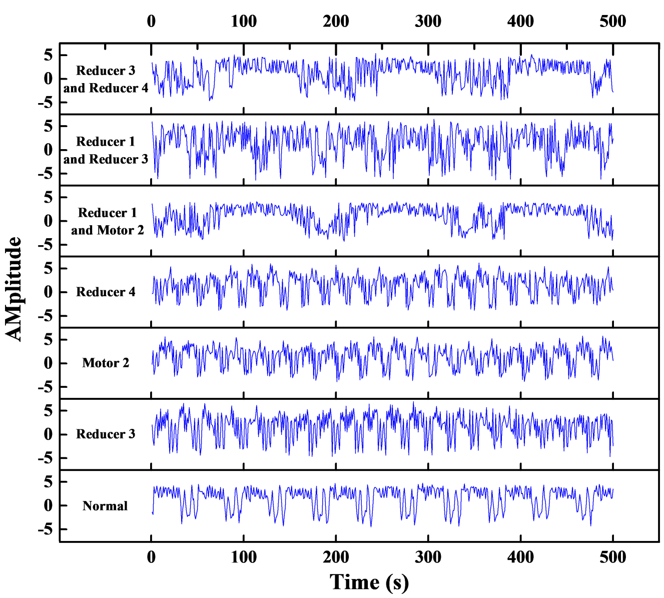



Fig. 4: Signals of different faulty categories.

comprises seven distinct categories: three individual fault types, three composite fault types, and a normal operational category. For this study, multi-hot encoding was adopted to label the data. Multi-hot encoding is an extension of one-hot encoding, designed to handle datasets where samples can belong to multiple categories simultaneously. This method assigns a binary value to each category in a vector, which facilitates the analysis and modeling of complex, multi-fault scenarios.

### *B. Setup for Benchmarking Experiment*

In this study, four experiments were conducted to evaluate the performance of our proposed model. The first experiment is a benchmarking test designed to assess the competitiveness of our model. For this purpose, we compared our model against eight different benchmark algorithms, ensuring that all models were configured with the same parameters for a fair comparison. The details of these benchmark algorithms are provided below:

1.**GCN** [**kipf2016semi**]: a semi-supervised graph convolutional network, which learns node representations by aggregating neighbor information.

2.**GAT** [**velivckovic2017graph**]: a graph neural network that uses attention mechanisms to weigh the importance of different neighbors, enhancing the model's ability to focus on relevant information.

3.**GIN** [**xu2018powerful**]: a graph neural network that matches the discriminative power of the Weisfeiler-Lehman graph iso-morphism test by using MLPs and sum aggregations in its architecture.

4.**SlimG** [**yoo2023less**]: a simple yet effective graph neural network designed for semi-supervised node classification, which combines linear feature transformations, structural features, and two-hop neighbor aggregation to achieve high accuracy, robustness, and interpretability, while being fast and scalable to large graphs.

5.**HGNN** [**feng2019hypergraph**]: a neural network framework designed for hypergraphs, capable of handling higher-order relationships through a hyperedge-based message passing scheme.

6.**HGNN+** [**gao2022hgnn+**]: an advanced hypergraph neural network that introduces adaptive hyperedge weighting and improved feature aggregation to enhance the modeling of complex interactions.

7.**HyperGCN** [**yadati2019hypergcn**]: a graph convolutional network specifically designed for semi-supervised learning on attributed hypergraphs, directly processing hypergraph data without conversion.

8.**TDHNN** [**zhou2023totally**]: dynamic hypergraph neural network that can dynamically adjust both the hypergraph structure and the number of hyperedges, using a trainable hyperedge feature distribution and attention mechanisms to optimize the hypergraph construction and improve representation learning.

In this study, these models are designed for fault diagnosis. The model employed multi-hot labels and used the Binary Cross-Entropy with Logits Loss as the loss function. The training process was set as 500 epochs, and the Adam optimizer was used with a learning rate of 0.001. For data preprocessing, the window sampling length was set to 1024. Data preprocessing involved k-means clustering (k=2) for feature-rich cluster selection and $k$-Nearest Neighbors ($K$-NN) hypergraph construction (k=15 neighbors). The feature convolution module utilized patch length 8 and stride 4 configurations, while the Progressive Hypergraph Structure Learning (PHSL) module operated with 2 celebrity node selection layers at 0.7 retention ratio. All tests were conducted on the Ubuntu 20.04

server featuring an Intel i9-10920X 3.50Ghz CPU and an Nvidia GeForce RTX 3090 graphics card.

### C. Setup for White Noises Experiment

A white noise experiment was conducted to evaluate the anti-noise capability of our proposed model. Different levels of Gaussian white noise were added to the raw current signals to simulate the noise from the environment, assets, and rotating components. In this study, the noise levels ranged from 30 dB to -30 dB and were divided into 8 different levels to comprehensively evaluate the anti-noise capability of the model. Performance was benchmarked against the baseline HGNN under identical noise-corrupted conditions.

### D. Setup for Ablation Experiment

To investigate the contributions of the two main modules in our model—Feature Convolution and Progressive Hypergraph Structure Learning— an ablation experiment was conducted. The experiment used a baseline model that lacked both the Feature Convolution and Progressive Hypergraph Structure Learning modules. The performance of the model was evaluated based on its accuracy to determine the impact of each module.

### E. Setup for Clustering Algorithms Experiment

To enhance the model's performance, a clustering mechanism was introduced in the pre-processing stage to extract useful and closely related features. This mechanism also aims to compress the feature length, accelerate model training, and lower the training cost. In this experiment, four clustering methods are compared to determine which is most effective for our mode. To evaluate the effectiveness of each clustering method, we measure the training time required to complete the training process and accuracy. The details of these clustering methods are as follows:

**1. Kmeans**: a centroid-based clustering algorithm that partitions the data into a specified number of clusters, minimizing the within-cluster variance. It is simple and computationally efficient but can be sensitive to the initial placement of centroids.

**2. Kmeans++**: an enhanced version of K-means that uses a more intelligent initialization method for centroids. This reduces the likelihood of suboptimal clustering and leads to better convergence and stability.

**3. DBSCAN**: a density-based clustering algorithm that groups together points that are closely packed together, marking points in low-density regions as outliers. It does not require the number of clusters to be specified beforehand and can discover clusters of arbitrary shape.

**4. Kshape**: a clustering algorithm specifically designed for time series data. It uses a shape-based distance measure to cluster time series based on their shape similarity, making it suitable for datasets with temporal patterns.

## V. Experimental Results

### A. Results of Benchmarking Experiemnt

The results of the benchmarking experiment are shown in Fig 5. The proposed model achieves a 96.81% accuracy, surpassing all benchmarks. It outperforms SlimG by 1.4% and TDHNN by 2.12%, demonstrating the effectiveness of integrating feature convolution and progressive hypergraph structure learning (PHSL). Traditional GNNs, such as GCN and GAT, exhibit significantly lower accuracy due to their inability to model high-order interactions inherent in compound faults.HGNN and HyperGCN outperform GNNs but still suffer from static hypergraph initialization. Removing PHSL from our model degrades accuracy by 5.2%, confirming its necessity for adaptive structure learning. Figure 6 demonstrates the accuracy and loss curves of different algorithms. It suggests the training process of the proposed algorithm is more stable in comparison with other advanced HGNNs.

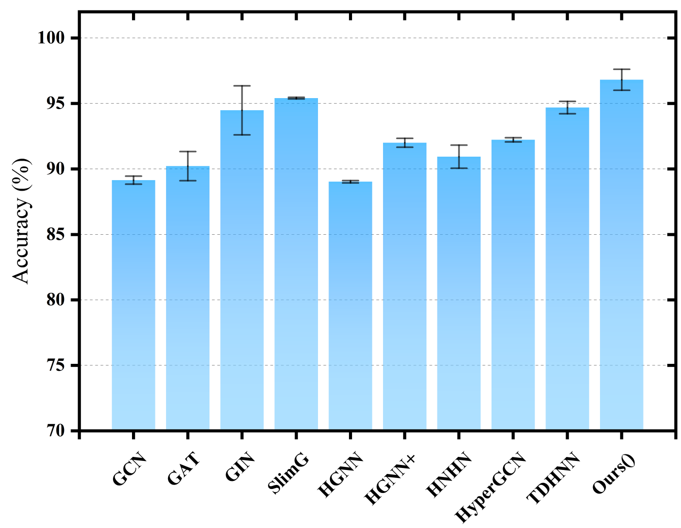


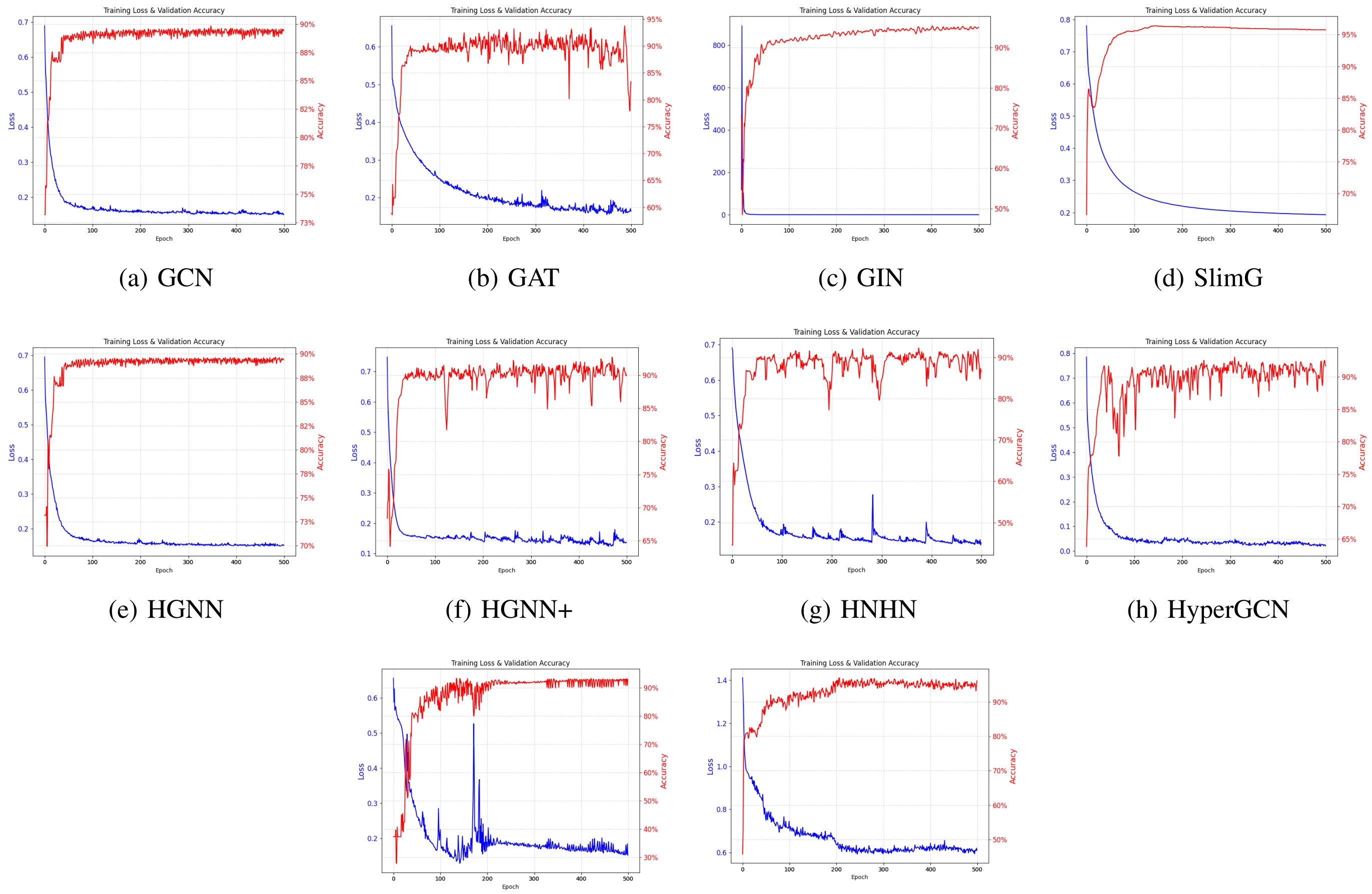

Fig. 5: Comparison of fault diagnosis accuracy for different algorithms.

### B. Results of White Noise Experiment

The results of the White noise experiment are reported in Fig 7. The proposed PHSL-based model maintains 95.85% accuracy at 10 dB and 82.25% accuracy at -5 dB, significantly outperforming HGNN. As shown in Fig. 7, the experimental results show that the model's performance begins to degrade when the noise level reaches -5 dB. Specifically, the training accuracy remains constant at this noise level, indicating a failure to learn effectively from the noisy data. At the -5 dB noise level, the model fails to correctly identify any of the multi-hot labels. Instead, it predicts all bits in the multi-hot labels as 0, leading to a constant accuracy value. This suggests that the noise level of -5 dB or lower severely disrupts the model's ability to distinguish between different fault conditions.

### C. Results of Ablation Experiment

The Full Model achieves 96.81% accuracy, outperforming the Baseline (89.03%). The ablation experiments validate the



(a) GCN (b) GAT (c) GIN (d) SlimG

(e) HGNN (f) HGNN+ (g) HNHN (h) HyperGCN

(i) TDHNN (j) Ours

Fig. 6: The accuracy and loss curves of different algorithms.

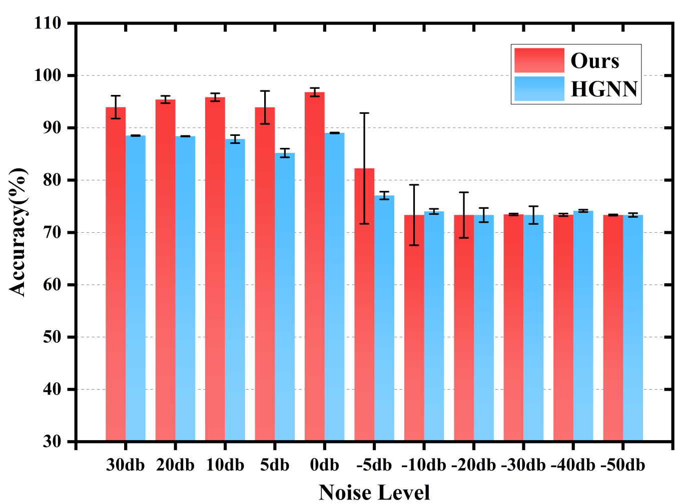



Fig. 7: The anti-noise performance of our model and HGNN algorithms in accuracy under data with different Gaussian white noise levels.

necessity of both FC and PHSL modules. The FC module suppresses noise and preserves discriminative features, while PHSL dynamically optimizes hypergraph structures to capture high-order fault correlations. Their integration elevates accuracy by 7.78% over the Baseline and establishes robustness against noise and structural ambiguity. The results of these ablation experiments are illustrated in Table I.

TABLE I: Results of Ablation Experiment

| **Condition** | **Accuracy** |
|---|---|
| With conv and HGSL | $96.81\% \pm 0.80\%$ |
| Without Conv | $90.22\% \pm 0.11\%$ |
| Without HGSL | $93.34\% \pm 1.87\%$ |
| HGNN (baseline) | $89.03\% \pm 0.09\%$ |

### D. *Results of Different Clustering Algorithms*

The experimental results are demonstrated in Fig 8. It can be seen that most of the clustering algorithms achieve higher clustering accuracy and less computational cost than the baseline model (94.70% ±0.0204). Specifically, the proposed model with K-means-based hypergraph initialization achieved the highest downstream classification accuracy of 96.81% (±0.0080), outperforming DBSCAN (95.17% ±0.0118), Kshpae (94.5% ±0.009), and Kmeans++ (95.75% ±0.0077). Computational efficiency further highlighted the advantage of Kmeans. It completed clustering in 197.83, significantly faster than Kshape (274.22s ±0.5633) and Kmeans++ (242.31s ±0.5864).

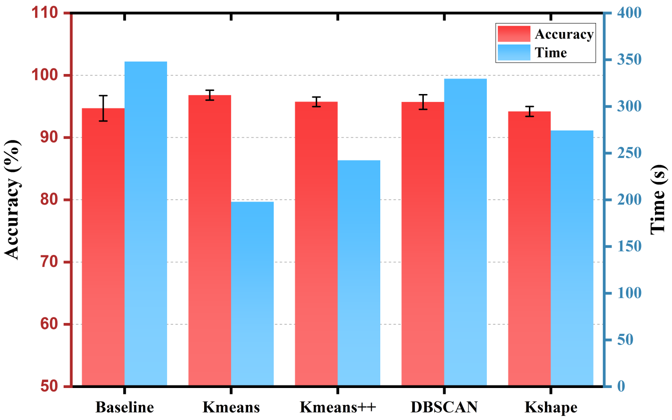



Fig. 8: The training time consumption and accuracy of different clustering methods.

## VI. Discussion

The experimental results demonstrate the effectiveness and reliability of the proposed industrial robot fault diagnosis algorithm. Integrating the Feature Convolution and PHSL modules achieves better diagnostic accuracy compared to state-of-the-art benchmarks. The Feature Convolution module demonstrates its importance by efficiently collecting local and global data while minimizing noise to improve features quality. The PHSL module achieves adaptive learning of complex high-order data correlations through its dynamic hypergraph structure adjustment mechanism based on node splitting and refinement. The combination of these improvements leads to enhanced method performance especially when dealing with noisy conditions which makes it suitable for industrial applications.

Our method shows merits in robotic fault diagnosis, but it faces several constraints. Real-time fault diagnosis systems might face difficulties because of the high computational complexity that exists within the PHSL module. The present model shows effective performance on seven fault categories but its ability to scale up for additional fault types remains unknown. The Feature Convolution module requires optimized performance across different operational environments because its performance depends on specific hyperparameters like patch length and stride. The proposed algorithm demonstrates solid performance on seven fault categories but additional enhancements are required to achieve maximum versatility and operational efficiency.

In the future, several directions can address these limitations and extend the capabilities of our proposed algorithm. Firstly, enhancing the efficiency of the PHSL module through optimization techniques or parallel computing strategies can reduce computational demands, which makes it more suitable for real-time applications. Moreover, Expanding training datasets to encompass diverse fault scenarios for improved generalization and robustness. Introducing time series large language model for data augmentation can be an effective way to address the few-shot issue. Another promising avenue involves exploring transfer learning methodologies to adapt the model to new operational contexts with minimal retraining. Addressing these areas is anticipated to yield a more versatile and efficient fault diagnosis system, ensuring safe and reliable industrial robot operation across diverse environments.

## VII. Conclusion

This paper introduces an industrial robot fault diagnosis algorithm, which contains three essential modules including Feature Convolution module, PHSL module, and HGNN module to deliver diagnostic results. The Feature Convolution module achieves better data quality through its ability to combine local and global information while reducing noise. The PHSL module performs dynamic hypergraph structure refinement to identify complex high-order data relationships accurately. The HGNN module executes fault diagnosis through an optimized hypergraph structure. The experimental study shows that the proposed method outperforms standard benchmarking algorithms. Our method delivers practical advantages that demonstrate its capacity to boost operational safety and reliability in industrial robots. Future research will concentrate on maximizing computational speed while making the model scalable to detect various faults and examining its applicability across different industrial environments.

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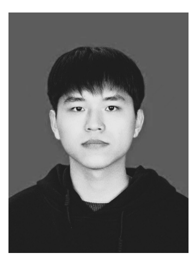

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